

Manufacturing Operational Insights Report *(Energy Document Chat (RAG))*

Course Name: Data Analytics

Institution Name: Medicaps University – Datagami Skill Based Course

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Academic Year: 2025-2026

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features *(Just mention key features here no need to go into details)*
2. Overall Architecture / Workflow
3. Tools & Technologies Used *(If applicable)*

Results & Output

Add the below details here:

1. *Screenshots / outputs*
2. *Reports / dashboards / models*
3. *Key outcomes*

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

Problem Statement

Energy-sector professionals rely heavily on large and complex documents such as technical reports, policy papers, equipment manuals, and research PDFs. Extracting precise information from these documents is time-consuming and inefficient using traditional keyword search or manual reading. Existing AI chat systems often provide generic or hallucinated answers that are not grounded in the actual document content.

There is a need for an intelligent system that allows users to upload energy-related documents and interact with them through accurate, context-aware question answering, ensuring that all responses are strictly derived from the provided documents.

Project Objectives

1. To design and implement a **Retrieval-Augmented Generation (RAG)** system for energy document intelligence.
2. To enable users to **upload and process PDF documents** related to the energy domain.
3. To perform **text extraction, chunking, and vector embedding** for efficient semantic retrieval.
4. To store embeddings in a **high-performance Vector Database** for fast and accurate similarity search.
5. To integrate an **LLM API** for generating precise answers grounded in retrieved document context.
6. To ensure that responses are **domain-specific, reliable, and free from hallucinations**.
7. To provide an intuitive **chat-based interface** for querying energy documents

Scope of the Project

- Uploading and processing of **energy-related PDF documents** (reports, manuals, guidelines).
- Text extraction and **semantic chunking** of documents.
- Generation of **vector embeddings** using modern embedding models.
- Storage and retrieval of embeddings using a **Vector Database**.
- Context-based question answering using **RAG architecture**.
- Responses strictly limited to the **content of uploaded documents**.
- Backend implementation using **Python**.
- Integration with **LLM APIs** for natural language interaction.

Proposed Solution

The proposed solution is a **Retrieval-Augmented Generation (RAG)–based conversational system** designed specifically for energy-domain documents. The system allows users to upload PDF documents, which are processed into semantically meaningful representations and stored in a vector database. When a user asks a question, the system retrieves the most relevant document chunks and uses a Large Language Model (LLM) to generate accurate, context-aware answers strictly grounded in the uploaded content. This approach ensures reliability, domain relevance, and reduced hallucinations.

Key Features

- PDF document upload and ingestion
 - Energy-domain focused document processing
 - Text extraction and intelligent chunking
 - Vector-based semantic search
 - Context-aware question answering
 - Responses grounded strictly in uploaded documents
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Overall Architecture / Workflow

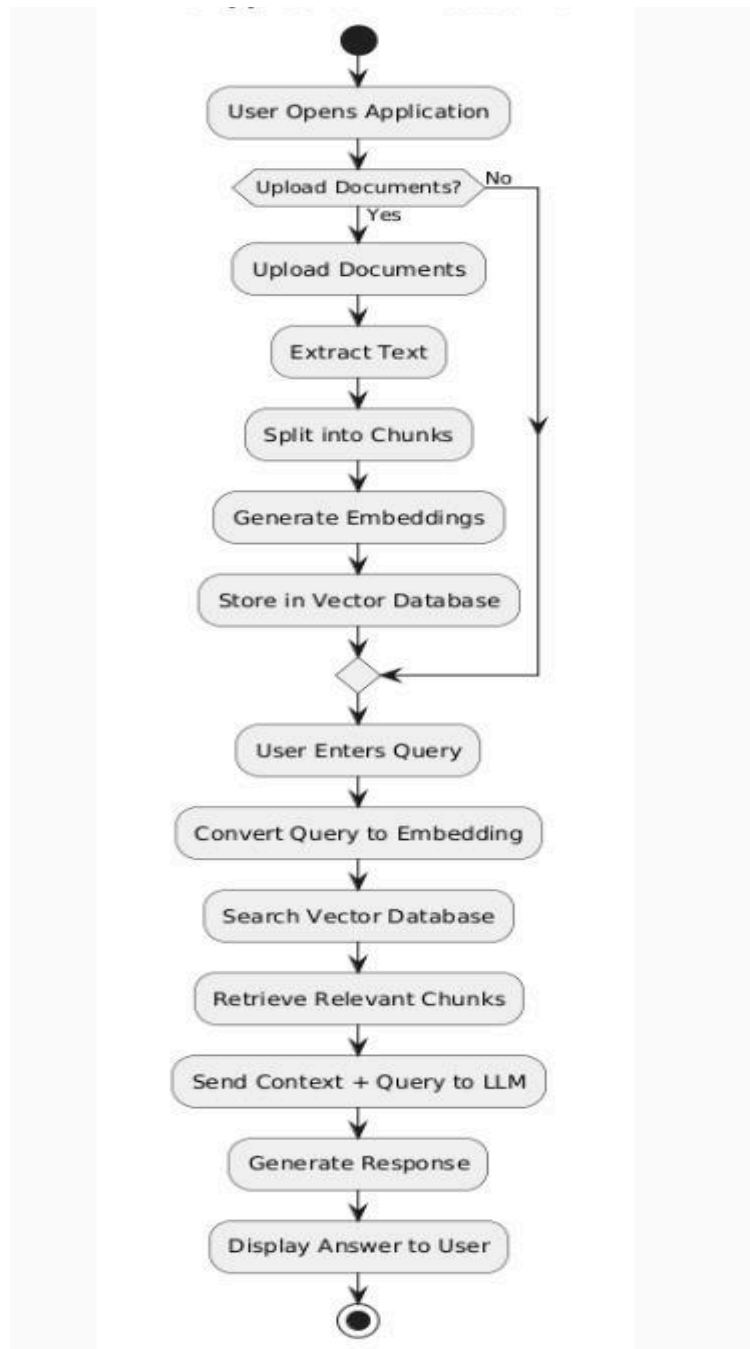


Fig: showing Workflow

1. Document Upload

Users upload energy-related PDF documents through the interface.

2. Document Processing

Uploaded PDFs are parsed to extract raw text, which is then cleaned and split into manageable chunks.

3. Embedding Generation

Each text chunk is converted into a vector embedding using an embedding model.

4. Vector Storage

The generated embeddings are stored in a Vector Database for efficient similarity search.

5. User Query

The user submits a natural language question via the chat interface.

6. Semantic Retrieval

The system retrieves the most relevant document chunks by comparing query embeddings with stored vectors.

7. Answer Generation

Retrieved context is passed to the LLM, which generates a precise answer grounded in the document data.

8. Response Delivery

The final answer is displayed to the user in a conversational format.

Tools & Technologies Used

- **Programming Language:** Python
- **Document Processing:** PDF text extraction libraries
- **Embedding Models:** LLM-based embedding APIs
- **Vector Database:** FAISS / Chroma
- **LLM API:** OpenAI / Groq
- **Backend Framework:** Python-based application framework
- **Architecture Pattern:** Retrieval-Augmented Generation (RAG)
- **Frontend Framework:** streamlit Python-based framework

Purpose	Model
Document embeddings	text-embedding-3-small
Query embeddings	text-embedding-3-small
Chat / Answering	gpt-4o-mini

Conclusion

The *Chat with Energy Documents (RAG)* project successfully demonstrates the design and implementation of a Retrieval-Augmented Generation system tailored for energy-domain document intelligence. The solution enables efficient ingestion, semantic indexing, and context-aware querying of professional energy documents such as reports and manuals. By combining vector databases with Large Language Models, the system delivers accurate, reliable answers that are strictly grounded in the uploaded content, significantly reducing hallucinations.

Through this project, key learnings include a strong understanding of RAG architecture, document chunking strategies, vector embeddings, semantic search, and effective integration of LLM APIs. The project also provided hands-on experience in building scalable AI pipelines, improving domain-specific information

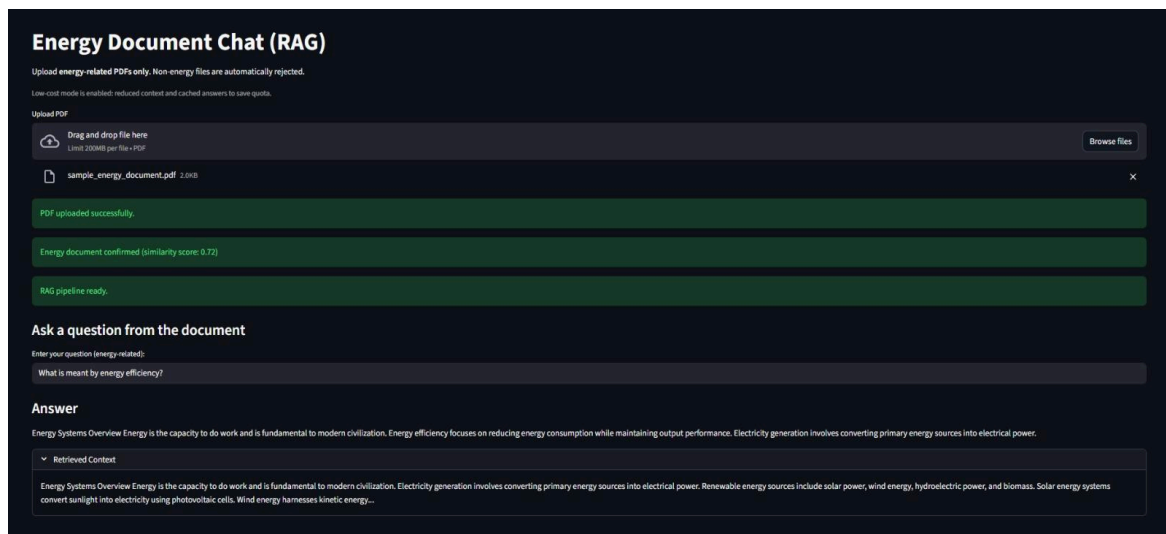


Fig:Streamlit-based user interface

“Interactive Energy Document RAG system enabling PDF-based question answering with semantic retrieval and context-grounded responses.”

<https://rag-1-qes6.onrender.com>

Future Scope & Enhancements:

1. Advanced Retrieval Techniques

- Upgrade from **FAISS Flat Index** to **HNSW or IVF indexes** for faster and more scalable vector search.
- Introduce **Hybrid Retrieval** by combining:
 - Dense embeddings (semantic search)
 - Sparse keyword search (BM25)
- Implement **query expansion** to improve recall for technical energy terms.

2. Multi-Document & Knowledge

Base Support

- Enable querying across **multiple energy documents simultaneously**.
- Add **document metadata filtering** (year, author, energy type: solar/wind/nuclear).
- Introduce **persistent vector storage** using:
 - ChromaDB
 - Weaviate
 - Pinecone

3. Security & Compliance

- Implement **role-based access control** for enterprise usage.
- Add **document encryption** at rest and in transit.
- Ensure **GDPR-compliant data handling**.
- Introduce **audit logs** for query tracking.